# SIMATS ENGINEERING

## Assessment -CO3-AT2

*CO3: Analyze knowledge representation and reasoning using propositional logic, FOL,*

*inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

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| **Subject** | Artificial Intelligence |
| **Course Code** | CSA 1712 |
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### Declaration

*I, Vignesh Kumar K (Reg No: 192512375), certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

## MCQ Answer Key — All 30 Questions

*The correct option is shown in bold and highlighted; a brief one-line rationale follows each question.*

### Q1. Which of the following best describes a Logical Agent in AI?

a) An agent that reacts randomly to the environment

**b) An agent that uses a knowledge base and logical inference to decide actions** ✓ Correct Answer

c) An agent that only follows pre-defined scripts

d) An agent that learns from rewards

*Rationale: A logical agent maintains a knowledge base and uses inference to decide its actions.*

### Q2. In Propositional Logic, 'P → Q' is logically equivalent to:

a) P ∧ ¬Q

b) ¬P ∨ Q

c) ¬Q → ¬P only

**d) Both b and c** ✓ Correct Answer

*Rationale: ¬P ∨ Q is the direct material-implication equivalence, and ¬Q → ¬P (the contrapositive) is also logically equivalent to P → Q — so both hold.*

### Q3. Conjunctive Normal Form (CNF) requires the formula to be expressed as:

a) A disjunction of conjunctions

**b) A conjunction of disjunctions (clauses)** ✓ Correct Answer

c) A conjunction of literals only

d) A sequence of implications

*Rationale: CNF is, by definition, an AND of ORs — a conjunction of disjunctive clauses.*

### Q4. In FOL, the sentence ∀x: Student(x) → Passes(x) means:

a) Some students pass

**b) Every entity that is a student passes** ✓ Correct Answer

c) If something passes, it is a student

d) There exists at least one student who passes

*Rationale: The universal quantifier applies the implication to every entity in the domain.*

### Q5. Unification in FOL is the process of finding substitutions that:

a) Convert clauses to CNF

**b) Make two logical expressions syntactically identical** ✓ Correct Answer

c) Prove a goal using backward chaining

d) Derive facts using Modus Ponens

*Rationale: Unification finds a substitution ϑ that makes two expressions match exactly.*

### Q6. In the Resolution algorithm, deriving the empty clause □ indicates:

a) The knowledge base is consistent

b) The negated goal is satisfiable

**c) A contradiction — the original goal is proven** ✓ Correct Answer

d) More clauses need to be added

*Rationale: Deriving □ means the negated goal led to a contradiction, proving the original goal true (proof by refutation).*

### Q7. Forward Chaining is best described as a ________ strategy.

a) Goal-driven / top-down

**b) Data-driven / bottom-up** ✓ Correct Answer

c) Heuristic-based depth-first

d) Probabilistic inference

*Rationale: Forward chaining starts from known facts and fires rules to derive new facts — a bottom-up, data-driven process.*

### Q8. Backward Chaining begins reasoning from:

a) All known facts toward a conclusion

**b) The goal and works backward to verify supporting facts** ✓ Correct Answer

c) A random clause in the knowledge base

d) The resolution refutation tree

*Rationale: Backward chaining starts at the goal and recursively reduces it to sub-goals matched against known facts.*

### Q9. Which step is NOT part of converting a sentence to CNF?

a) Eliminate biconditionals (↔)

b) Move negations inward using De Morgan's laws

c) Apply Skolemization to remove existential quantifiers

**d) Distribute AND over OR** ✓ Correct Answer

*Rationale: CNF conversion distributes OR over AND (to get a conjunction of disjunctions) — 'distribute AND over OR' describes the wrong direction and is not a genuine CNF step.*

### Q10. Knowledge Engineering in FOL involves:

a) Training a neural network on domain-specific data

**b) Defining domain concepts, relations, and axioms formally in FOL** ✓ Correct Answer

c) Writing heuristic search strategies

d) Encoding reward functions for reinforcement learning

*Rationale: Knowledge engineering is the process of formally encoding a domain's concepts, relations, and axioms in FOL.*

### Q11. Modus Ponens states: If P → Q is known and P is true, then:

a) ¬Q is true

**b) Q must be true** ✓ Correct Answer

c) P is uncertain

d) The KB is inconsistent

*Rationale: Modus Ponens: from P → Q and P, infer Q.*

### Q12. The Most General Unifier (MGU) is the substitution that:

a) Is the most specific possible match

**b) Makes two terms identical with minimal variable binding** ✓ Correct Answer

c) Eliminates all variables from a clause

d) Applies resolution to produce CNF

*Rationale: The MGU is the least-constraining substitution that still unifies the two terms.*

### Q13. Which inference method is more appropriate when a specific goal is known?

a) Forward Chaining

b) Truth Table method

**c) Backward Chaining** ✓ Correct Answer

d) A* search

*Rationale: Backward chaining works efficiently backward from a known goal to check supporting facts, avoiding irrelevant derivations.*

### Q14. Propositional Logic differs from First Order Logic because it:

a) Supports probabilistic reasoning

**b) Cannot represent objects, relations, or quantifiers — only truth values of atomic propositions** ✓ Correct Answer

c) Is always more expressive

d) Handles uncertainty better

*Rationale: Propositional logic only deals with atomic true/false statements, lacking FOL's objects, predicates, and quantifiers.*

### Q15. Resolution Refutation proves a goal G by:

a) Directly deriving G from the KB using forward chaining

**b) Adding ¬G to the KB and deriving a contradiction (empty clause □)** ✓ Correct Answer

c) Checking G in the truth table

d) Applying Modus Tollens repeatedly

*Rationale: Refutation proves G by assuming ¬G, combining with the KB, and showing this leads to a contradiction (□).*

### Q16. Learning from observation in AI refers to:

a) An agent receiving numerical rewards from the environment

**b) Improving performance by examining examples or experience from the environment** ✓ Correct Answer

c) Acquiring knowledge by following explicit symbolic rules

d) Memorizing a fixed dataset without generalization

*Rationale: Learning from observation means improving future performance based on past examples/experience.*

### Q17. Inductive learning is best described as:

a) Deriving specific conclusions from general rules

**b) Generalizing a hypothesis from specific training examples** ✓ Correct Answer

c) Memorizing training data exactly

d) Applying search algorithms to find optimal policies

*Rationale: Inductive learning generalizes from specific labelled examples to a broader hypothesis.*

### Q18. In a Decision Tree, what does each internal node represent?

a) A class label / output decision

b) A probability distribution over outcomes

**c) A test on an attribute / feature** ✓ Correct Answer

d) A training data sample

*Rationale: Internal nodes test an attribute/feature; leaf nodes hold the class label/decision.*

### Q19. Explanation-Based Learning (EBL) differs from inductive learning because it:

a) Requires large amounts of training data to generalize

b) Learns only from numeric data using regression

**c) Uses prior domain knowledge to explain and generalize a single example** ✓ Correct Answer

d) Applies unsupervised clustering to find hidden patterns

*Rationale: EBL uses existing domain theory to explain one example and generalize a rule from it, needing far less data than pure induction.*

### Q20. In statistical learning, a Naive Bayes classifier assumes:

a) All features are conditionally dependent on each other given the class

b) The data must follow a uniform distribution

**c) All features are conditionally independent given the class label** ✓ Correct Answer

d) Labels are determined by a reward function

*Rationale: The 'naive' assumption is conditional independence of features given the class label.*

### Q21. Learning with complete data (no hidden variables) in a Bayesian network involves:

a) Using the EM algorithm to estimate missing values

b) Training a deep neural network with dropout regularization

**c) Directly computing maximum likelihood estimates from observed data** ✓ Correct Answer

d) Applying reinforcement signals to update weights

*Rationale: With complete data, parameters can be estimated directly via maximum likelihood — no EM needed.*

### Q22. The Expectation-Maximization (EM) algorithm is used when:

a) Training data is fully labeled and no variables are hidden

b) An agent explores an unknown environment using trial and error

**c) The dataset contains hidden (latent) variables that cannot be directly observed** ✓ Correct Answer

d) Backpropagation fails to converge in a neural network

*Rationale: EM is specifically designed to estimate parameters when some variables are hidden/latent.*

### Q23. In a Neural Network, the activation function serves to:

a) Initialize the weights of each neuron to zero

b) Normalize the input dataset before training

**c) Introduce non-linearity to enable learning of complex patterns** ✓ Correct Answer

d) Store previously learned examples in memory

*Rationale: Without a non-linear activation function, a network would collapse to a linear model regardless of depth.*

### Q24. Which of the following correctly describes backpropagation in a neural network?

a) Weights are updated in the forward pass using input gradients

b) Outputs of each layer are stored and reused in later epochs

**c) The error is propagated backward from output to input layers to adjust weights** ✓ Correct Answer

d) A separate model is trained for each hidden layer independently

*Rationale: Backpropagation computes the output error and propagates gradients backward through the layers to update weights.*

### Q25. A Support Vector Machine (SVM) — a type of kernel machine — classifies data by:

a) Minimizing the number of training examples used

b) Grouping similar data points into clusters using centroids

**c) Finding the maximum-margin hyperplane that separates classes** ✓ Correct Answer

d) Updating weights using Q-values from an environment

*Rationale: SVM finds the hyperplane that maximizes the margin between classes.*

### Q26. The kernel trick in kernel machines (SVMs) allows:

a) Reducing the number of support vectors needed

b) Accelerating gradient descent convergence

**c) Operating in a high-dimensional feature space without explicitly computing coordinates** ✓ Correct Answer

d) Replacing hidden layers in a deep neural network

*Rationale: The kernel trick computes inner products in a high-dimensional space implicitly, avoiding explicit coordinate transformation.*

### Q27. In Reinforcement Learning, the agent aims to maximize:

a) Classification accuracy on a fixed test set

b) The number of features selected from the training data

**c) Cumulative reward received over time through interaction with the environment** ✓ Correct Answer

d) The size of the neural network hidden layer

*Rationale: RL agents learn a policy that maximizes expected cumulative (long-term) reward.*

### Q28. The Q-Learning algorithm in Reinforcement Learning is classified as:

a) A supervised learning algorithm requiring labeled datasets

b) An unsupervised clustering technique using centroids

**c) A model-free, off-policy temporal difference learning method** ✓ Correct Answer

d) A decision-tree-based ensemble classifier

*Rationale: Q-Learning learns action-values directly from experience without a model of the environment, and off-policy since it learns about the greedy policy while following another.*

### Q29. Overfitting in machine learning occurs when:

a) The model performs poorly on both training and test data

b) The learning rate is set too low during gradient descent

**c) The model fits training data too closely and generalizes poorly to new data** ✓ Correct Answer

d) The training set has more features than training examples

*Rationale: Overfitting = excellent training performance but poor generalization to unseen data.*

### Q30. Which of the following best distinguishes Reinforcement Learning from Supervised Learning?

a) RL uses labeled input-output pairs; SL uses reward signals

b) Both RL and SL require explicit reward functions

**c) RL learns via trial-and-error with reward feedback; SL learns from labeled examples** ✓ Correct Answer

d) RL is always faster to converge than SL on any dataset

*Rationale: RL learns a policy through trial-and-error reward feedback, while SL learns a mapping from labeled input-output examples.*